

Blockchain: The future of Geospatial Applications

Blockchain can be regarded as the world's leading software platform for digital assets. It is a continually growing list of records, called blocks, which are linked and secured by using cryptography. A blockchain inherently resists the modification of data.



In the early stages of the Internet's evolution, computers required a specialised language like humans, in order to communicate. For instance, carpenters use certain terms that are well understood amongst them, just as doctors do or even sailors. Each group uses a shared language that is designed to enhance precision and clarity within its profession.

Computers use protocols to interact with one another. These protocols include HTTP, which is used to access the Internet through a Web browser; SMTP, which helps deliver emails to the right recipients; and SNMP, which is used for building a computer network.

These protocols are invisible to us and we only see the applications that take advantage of them. For instance, we use Internet Explorer or the Firefox browser to access websites, while Outlook provides access to our email. SNMP is virtually invisible to most of us unless we've worked on a computer network that's bigger than the basic router setup most of us use at home.

Enter blockchain

After the release of the bitcoin in 2009, blockchain slowly became an Internet protocol that transmitted value from node to node. One can think of blockchain as a way that allows different computers to talk to one another using the language of cryptography. It is a peer-to-peer network; thus the information exchanged does not require centralised servers. Since there is no central server that 'runs' a blockchain, it is

incredibly difficult to use the current denial-of-service attacks on it, which work on centralised applications.

A blockchain is a digital, decentralised, peer-to-peer, immutable, public, shared ledger/registry. Each transaction has a string of characters called a hash. Each hash includes a date/time stamp, a unique ID, a code linking it to the previous hash and a private key identifying ownership, albeit anonymously. Each transaction is another link in a chain that can be traced backwards to the previous link, all the way back to the origin of the entire chain, which is called the Genesis Block.

Unlike other databases or registries, a blockchain doesn't reside on a single computer and is not controlled by any person or any group of people. It resides on an open network of computers (also called nodes) where all nodes are equal – no super node manages the other nodes. Anyone can download the software and become a fully qualified node in the network, and all nodes carry a full copy of the blockchain registry. Data can only be entered or transferred with encrypted signatures using private encryption keys that only the owner of each particular data entry possesses. Nodes on a network record each transaction, and these transactions are mirrored on every other node throughout the network.

The transactions can be accessed and downloaded from any node on the network. This makes a blockchain an open and distributed ledger. When a node is out of sync with other nodes on the network, it is rejected until it is reset to match the other nodes. That makes it impossible to alter any records, making a blockchain immutable.